Low radioactivity techniques for gas detectors in rare events searches

Giulia D'Imperio - INFN Roma

DRD1 WG3 workshop - 30/10/2026

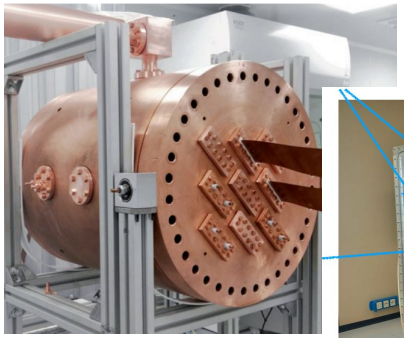
Gas detectors for rare events

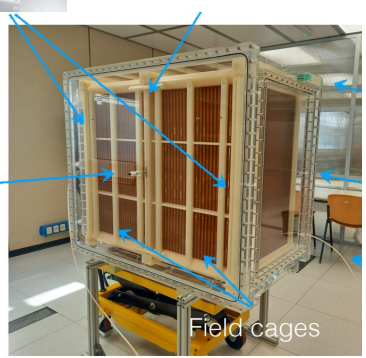
Gaseous TPCs are **versatiles detectors**
→ increasing interest for **rare events searches** (dark matter, double-beta decay, axions, ..)

- Good performance on **spatial and energy resolution**
- **Topological information** to discriminate signal from background
- **Directionality** of tracks
- Operation **stability** and homogeneity
- Possibility of **scaling-up**
- Potential **ultra-low background** conditions: made mainly of radiopure materials like Cu, plastics (kapton, PMMA, nylon)

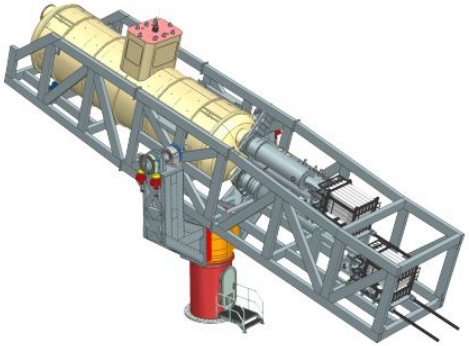
Low radioactivity is a key feature for experiments that search **rare events**

Examples rare events searches with TPCs





TREX-DM, CYGNO
for **dark matter**

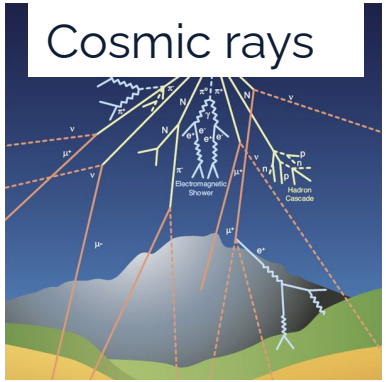


CAST, IAXO for **solar axions**



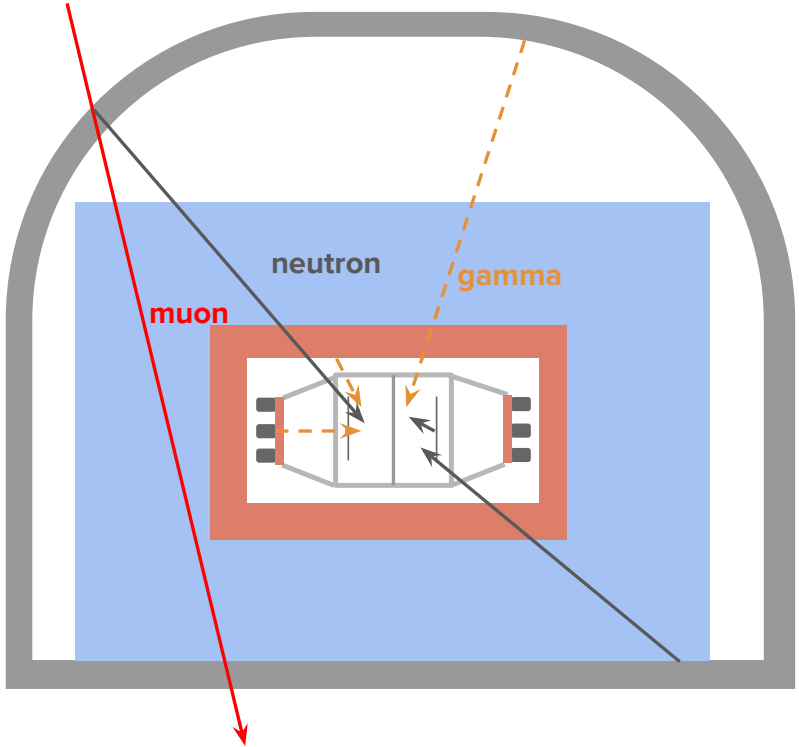
PANDAX for
neutrinoless double beta decay

Background sources



Radioactivity

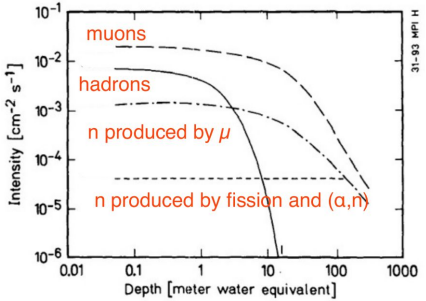


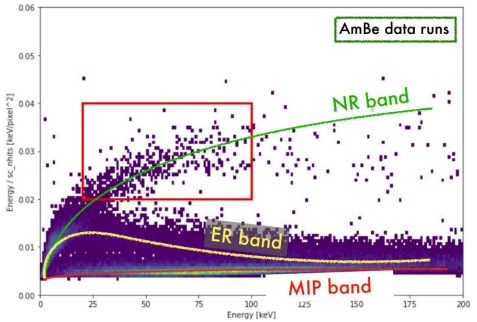


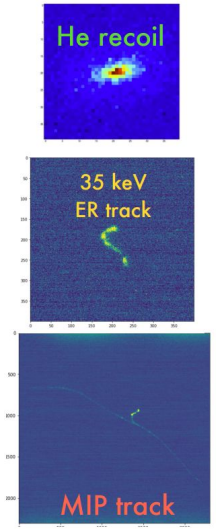
- External: cosmogenic neutrons (origin: muon interactions)
- External: radiogenic n/γ (origin: radioactivity of rocks and concrete of the lab)
- Radioactivity of the materials in setup, including cosmogenic activation: α, β, γ radiation, neutrons

Mitigation techniques

- **Deep underground facilities**
  - e.g LNGS under 1400 m of rock (3800 mwe)
- **Passive shielding**
  - Gamma: Pb, Cu
  - Neutrons: water, polyethylene
- **Active shielding**
  - Fiducial volume, Veto
- Control of **radiopurity of materials**
  - Screening, careful selection of materials
  - Material treatment: bulk, surface
  - Minimizing cosmogenic activation
- Techniques for **background rejection**
  - Topology of events (ER vs NR)
  - Directionality

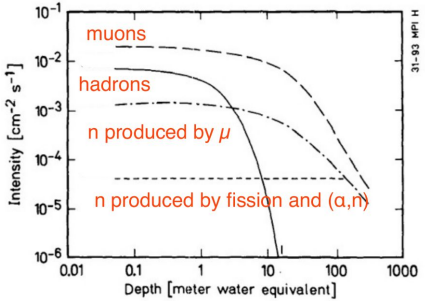


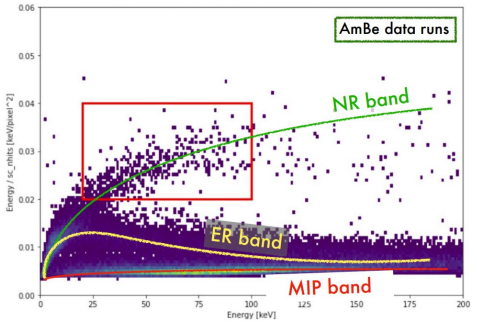


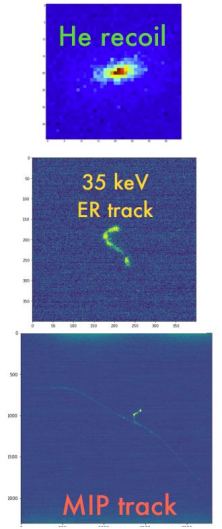


Mitigation techniques

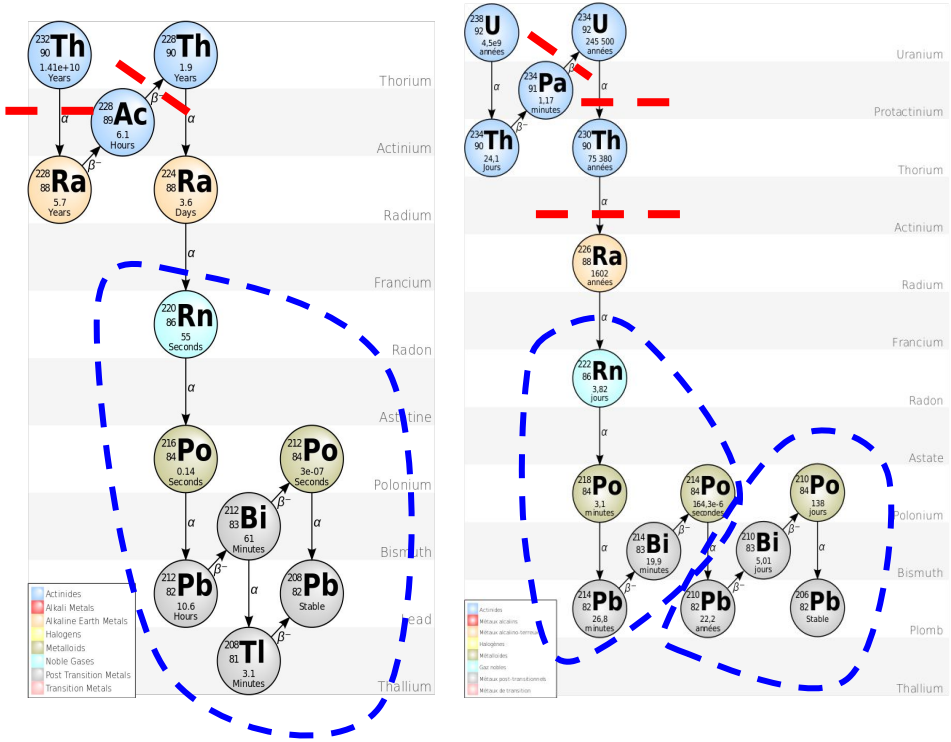
- **Deep underground facilities**
  - e.g LNGS under 1400 m of rock (3800 mwe)
- **Passive shielding**
  - Gamma: Pb, Cu
  - Neutrons: water, polyethylene
- **Active shielding**
  - Fiducial volume, Veto
- Control of **radiopurity of materials**
  - Screening, careful selection of materials
  - Material treatment: bulk, surface
  - Minimizing cosmogenic activation
- Techniques for **background rejection**
  - Topology of events (ER vs NR)
  - Directionality





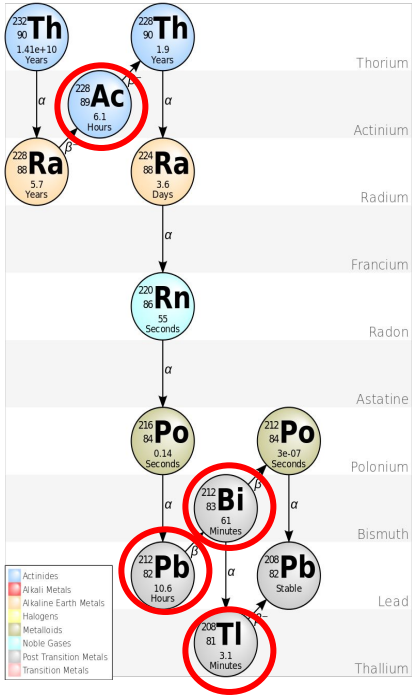


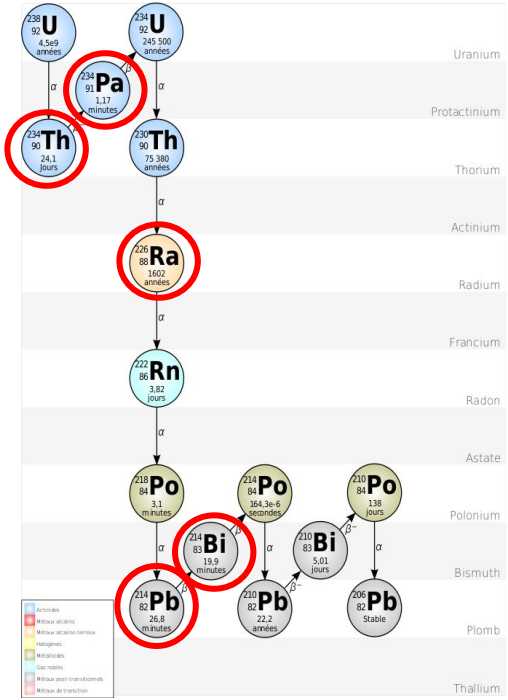
Radioactivity background



- Natural radioactivity: ^{238}U, ^{235}U, ^{232}Th and ^{40}K
- anthropogenic: ^{137}Cs
- Radon
- cosmogenically activated isotopes
- Secular equilibrium may be broken in correspondence to isotopes with long half-life
- Radon plate-out can also create sub-chains with different activities

Radioactivity screening: HPGe





- HPGe screening up to 10-100 µBq/kg precision for gamma emitters
  (for CYGNO done in STELLA facility at LNGS)

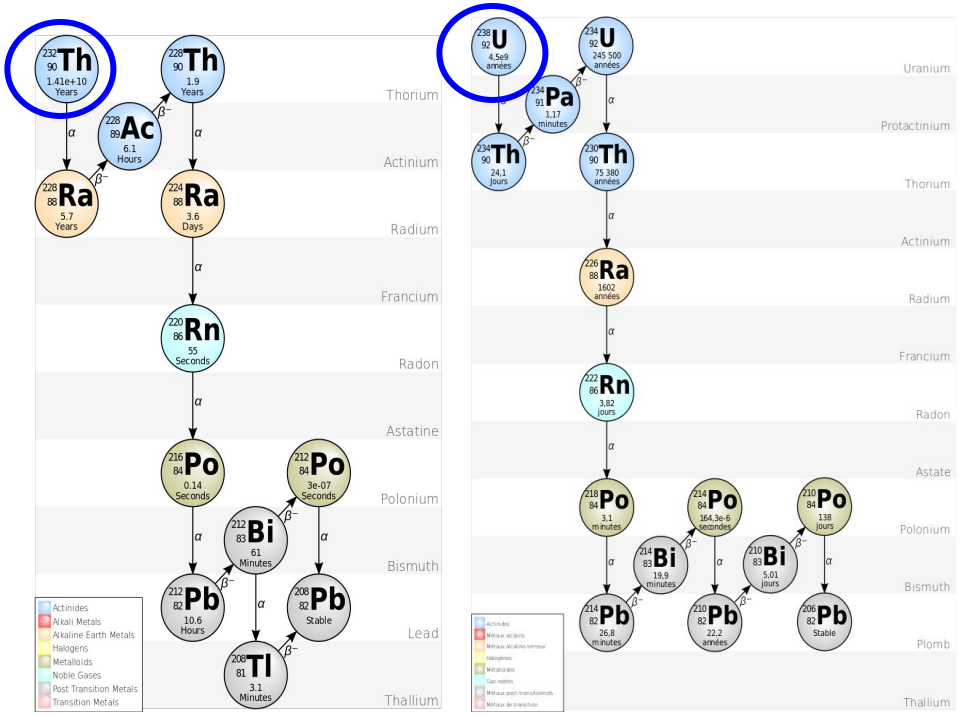


☺ Non-destructive: components to be actually used can be analyzed

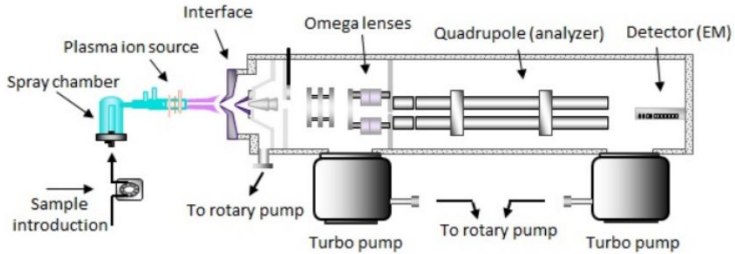
☺ Sensitive to particular radioactive isotopes

☹ Large samples, long measurements required

Radioactivity screening: ICP-MS



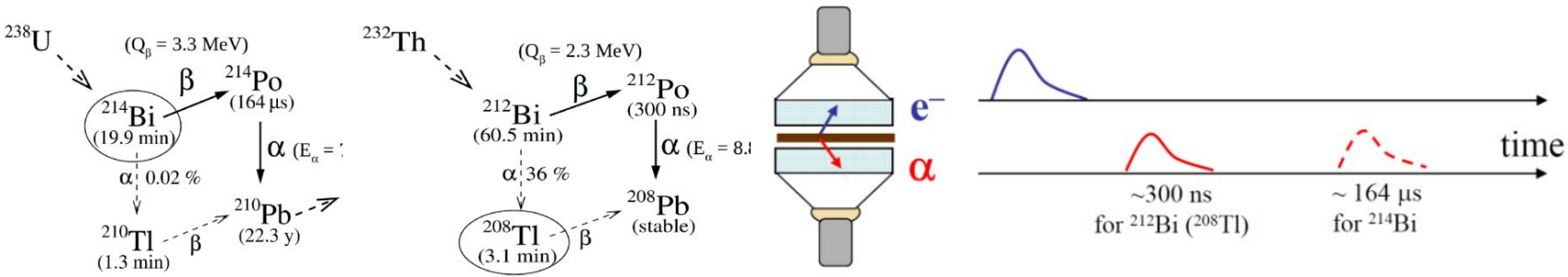
- ICP-MS can measure the concentration of primordial nuclides U, Th at ppt level
  → 1-10 μBq/kg precision
  (Chemistry lab service at LNGS)



☺ Fast: results provided within a few days
☺ Very small samples required
☹ Destructive

Other techniques

- Alpha spectroscopy
- Neutron Activation Analysis (NAA)
- Special detectors: Bi-Po detectors built by the SuperNEMO col. and operated at LSC
  - Sensitive to levels of 0.1 μBq/kg of the lower parts of 238U, 232Th through BiPo α-β coincidences in very thin samples (<100-200 μm thick)



Monte Carlo simulations

Monte Carlo simulations of the interaction of background radiation in matter are extremely useful to:

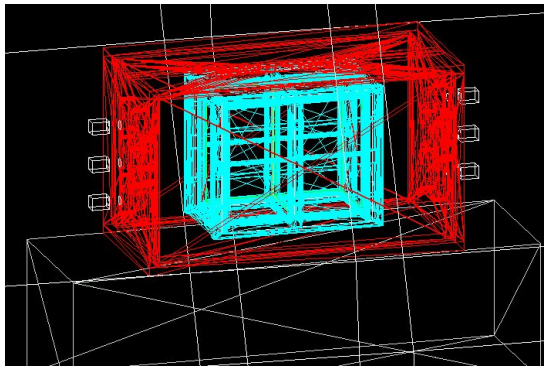
- Study the effect of **background reduction strategies** (shieldings, particle discrimination, …)
- Evaluation of **expected counting rates and sensitivity** of future experiments
- Understanding of measured data → **background models**

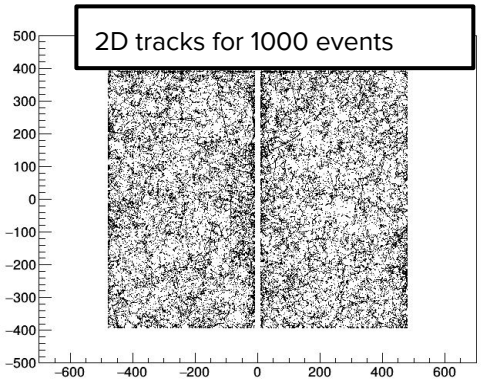
Many tools available → CYGNO choice:

- **Step 1: GEANT4**: define primary particles + physical processes + geometrical setup
- **Step 2:** Detector response with a custom code based on C++ (**digitization**)

CYGNO simulation: Geant4

- In each simulation: user defines the source of radioactivity
  - **external** → γ,n from a surface containing the full setup
  - **internal**
    → setup parts are simulated separately as “active” sources
    → each isotope is simulated individually and results are scaled for initial activity
- Energy deposits in the active gas are stored in the Geant4 output (x,y,z,dE)

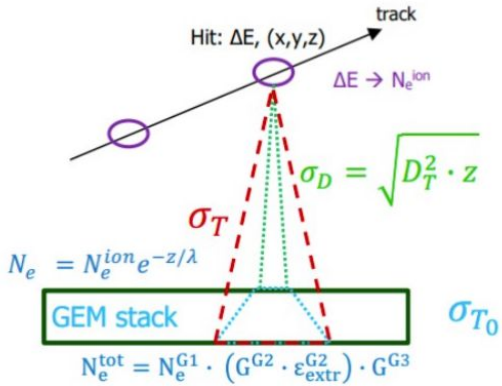




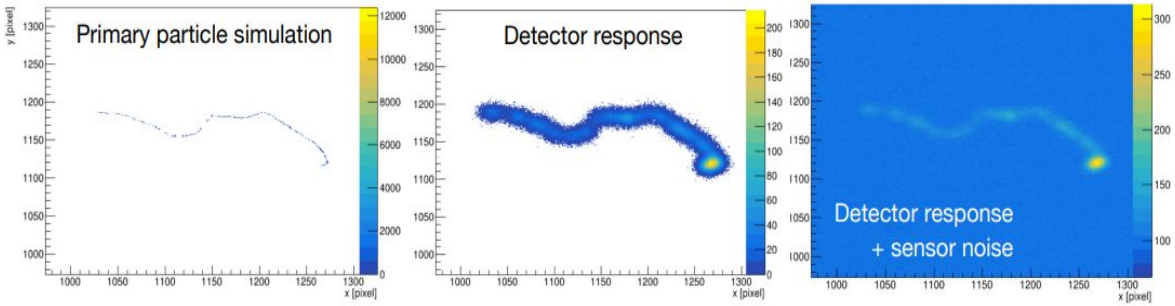
CYGNO simulation: “digitization”

The following processes are considered (including fluctuations):

- **Ionization**
- **Diffusion** (in the gas and in the GEMs)
- **Absorption** in the gas
- **Multiplication** in the 3-GEM stack
- **Gain Saturation** effect (depending on the charge density)
- **Light production** and collection
- Sensor **noise**

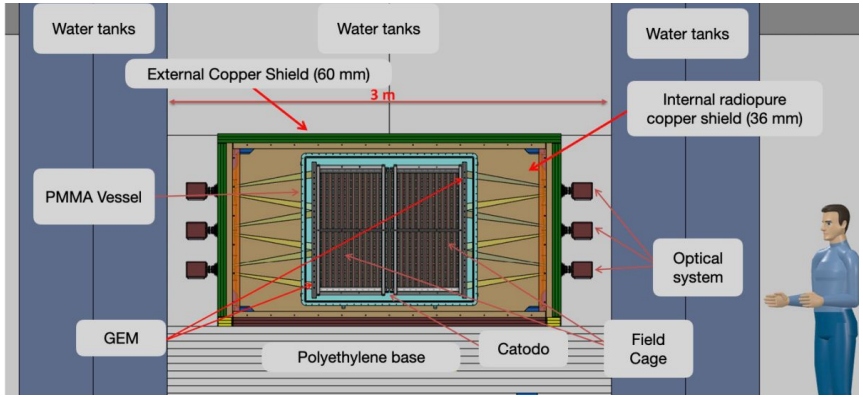


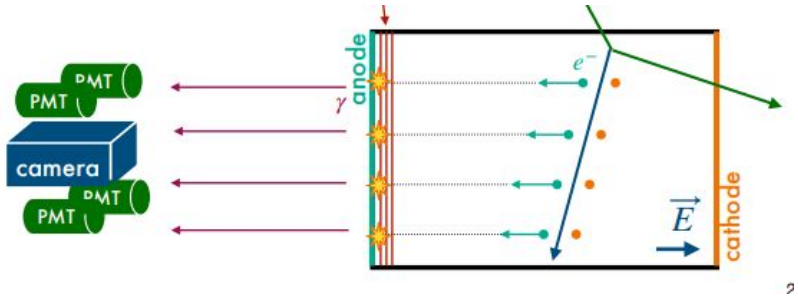
Output of digitization: same format as the real data



The CYGNO-04 detector

- TPC with **2 chambers** with a common cathode and **drift length 50+50 cm**
- **Field cage** made of kapton foil + Cu strips generating a field of 1 kV/cm
- **2 sets of triple GEMs** 50 cm x 80 cm operated at 440-460 V
- **Optical readout**: **3+3 cameras** with rectangular sensors (ORCA Quest) and **8+8 PMTs**
- **Vessel: low radioactivity PMMA** filled with He/CF4 (60/40) at atmospheric pressure
- **Shielding**: 10 cm copper + 100 cm water with a polyethylene base

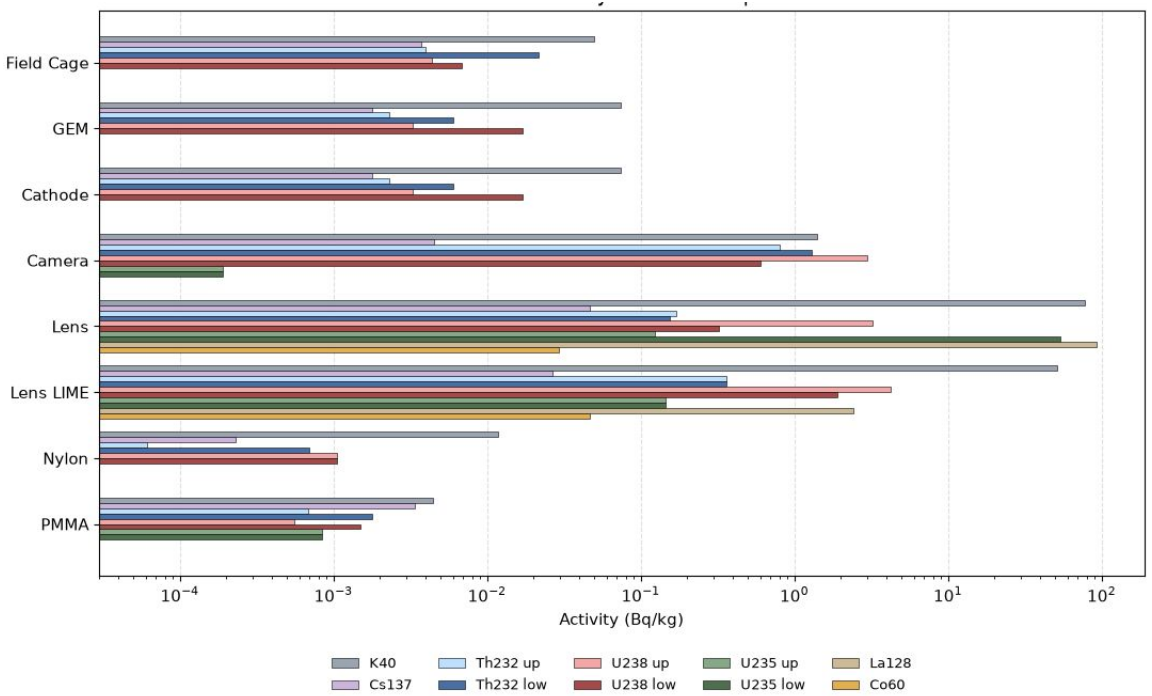




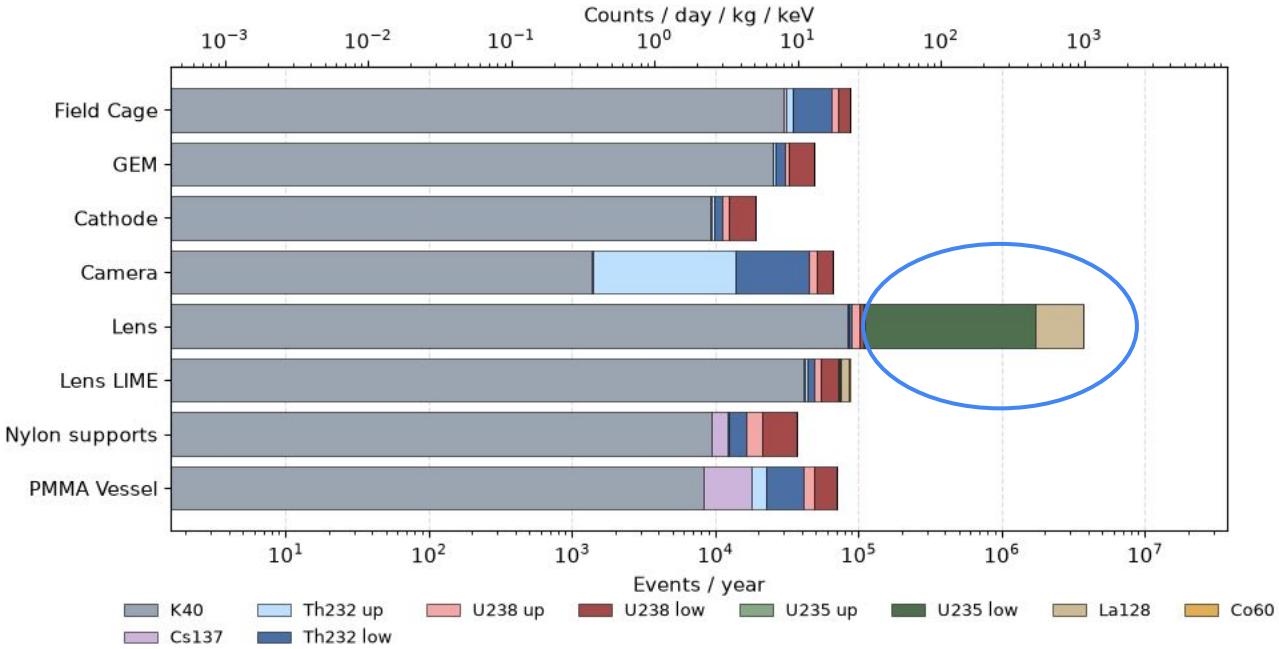
Now under commissioning at LNF overground → underground @LNGS by the end of 2026

CYGNO-04 materials screening

- Extensive campaign to screen all the materials and components used for CYGNO-04 detector
- All measurements done at LNGS with HPGe and ICP-MS



CYGNO-04 background

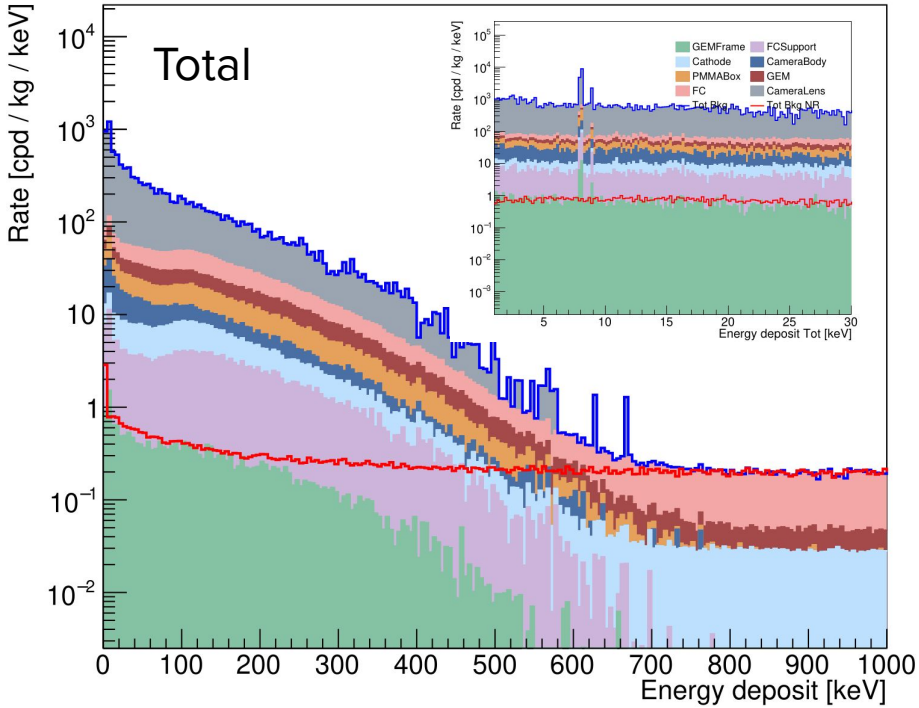


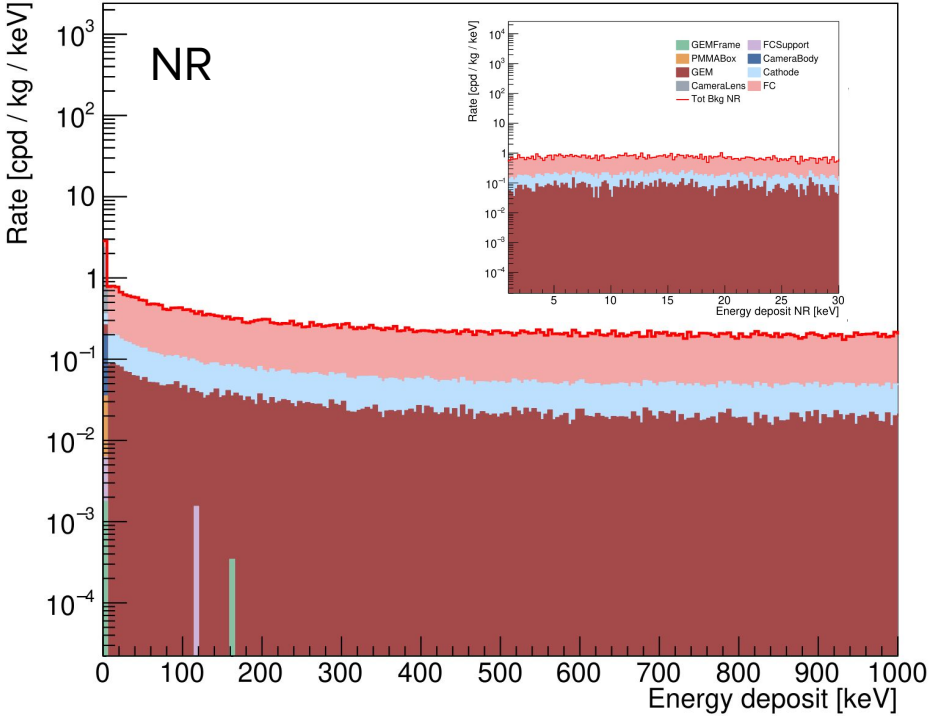
Lens used in LIME prototype: Schneider Xenon with 25.6 mm focal length and 0.95 aperture

CYGNO-04 lens: EHD imaging 25 mm focal length and 0.85 aperture
→ about 2x factor in light collection but large background

Estimated background from radioactivity of materials is of the order of **3 x 10^6 events/year** (about a factor x0.1 with Xenon lenses used in LIME)

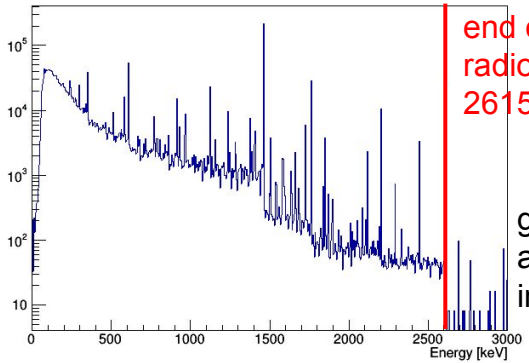
CYGNO-04 background spectrum





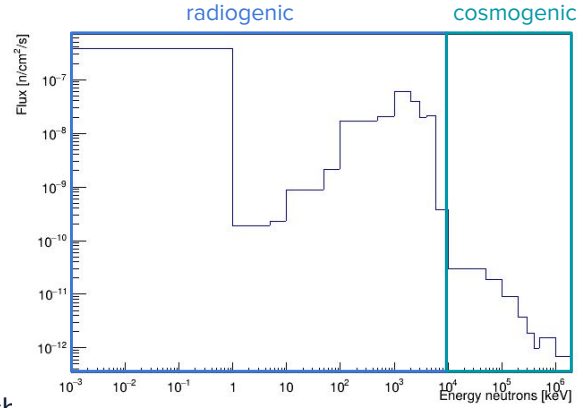
External background

- Ambient **gamma and neutron fluxes are site specific** and can change up to a factor 2 between experimental halls at LNGS
- Dedicated gamma measurement campaign in CYGNO-04 area with a NaI detector
  → Gamma flux **~0.5 gamma/cm²/s**
- Shielding (10 cm Cu + 100 cm water) gives an attenuation on gammas of about 10^5
  → expected gamma background **O(10^4) events/year**
- Neutron flux measurements are more complex, but LNGS flux at different energies is available in literature → expected neutron background is negligible



end of natural radioactivity spectrum 2615 keV line of ^{208}Tl

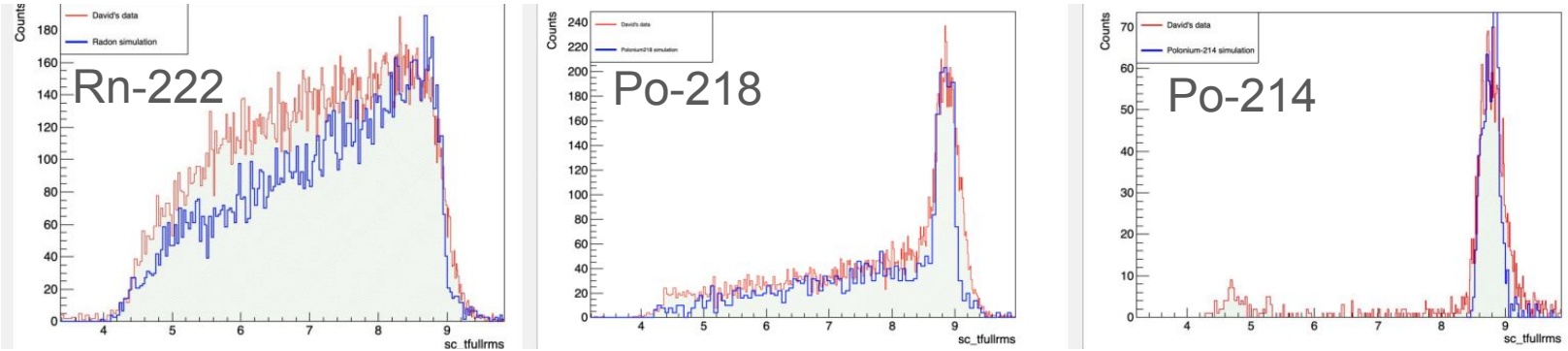
gammas from muons and neutrons interactions



Gaseous backgrounds

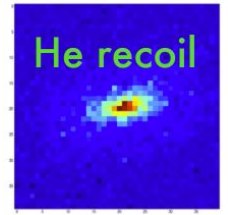
- CYGNO uses a gas mixture of He/CF4 (60/40)
  - ^{14}C should not be an issue as long as the carbon comes from hydrocarbon fuels
  - estimated background $<10^4$ events/year
- Radon:
  - Internal volume inside shieldings is flushed with N2 to avoid radon
  - LIME (20 L prototype operated underground 2022-2024) we observed evidence of radon

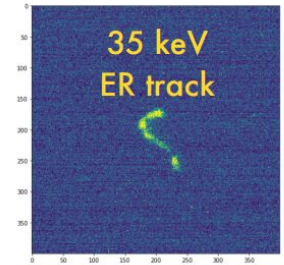
Reconstructed z position of alpha peaks in radon daughters (red data, blue MC)

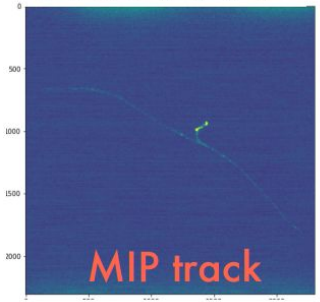


ER/NR discrimination

- Nuclear recoil (NR) and electron recoil (ER) tracks are very different for energies > 20 keV, **discrimination challenging at few keV**
- Simple cuts on track density: demonstrated **ER background rejection @5.9 keV of 96.5% (99.2%)** with 50% (40%) NR efficiency
- More sophisticated algorithms based on machine learning trained with Monte Carlo are under study
  → **ER rejection ~10^3-10^4 around 20 keV** with 50% NR efficiency







TREX-DM micromegas

Activity (in μBq/cm²) from HPGe and BiPo-3 detectors (factor F to convert to mBq/kg)

Sample	Measurement	^{238}U	^{226}Ra	^{232}Th	^{228}Th	^{235}U	^{40}K	^{60}Co	^{137}Cs	F
Cu-Kapton-Cu foil	BiPo-3		<0.038		<0.043					80.9
CAST MM	BiPo-3		<0.39		<0.20					33.9
MM#0	#0	<49	<0.70	<1.2	<0.35	<0.22	<2.3	<0.14	<0.13	21.5
	BiPo-3		<0.064		<0.016					21.5
MM#1	#0	<5.2	<0.10	<0.22	<0.08	<0.03	3.45±0.40	<0.02	<0.02	74.6
	#1	7.41±0.81	<0.21	0.19±0.05	<0.11	0.36±0.04	0.84±0.16	<0.02	<0.03	74.6
	#2	7.87±0.89	<0.17	0.14±0.04	0.07±0.02	0.36±0.04	0.81±0.15	<0.03	<0.02	74.6
MM#2	#0	<11	<0.28	<0.26	<0.17	<0.11	1.80±0.65	<0.06	<0.08	81.2
	#1a	<11	<0.25	<0.26	<0.15	<0.14	<1.1	<0.06	<0.07	81.2
	#1b	<4.1	0.140±0.017	<0.19	<0.068	<0.04	0.43±0.13	<0.03	<0.02	81.2
MM#3	#0	<2.5	0.035±0.004	<0.067	<0.018	0.022±0.004	0.102±0.030	<0.006	<0.008	81.2
MM#4	#0	<17	<0.19	<0.58	<0.14	<0.13	1.07±0.23	<0.03	<0.04	70.8
	#1	<12	<0.12	<0.49	<0.10	< 0.07	<0.63	<0.03	<0.03	70.8
	#2	<16	<0.13	<0.41	<0.14	< 0.10	<1.2	<0.03	<0.05	70.8
Vacrel	#0	<19	<0.61	<0.63	<0.72	<0.19	4.6±1.9	<0.10	<0.14	73.3
	BiPo-3		<0.037		<0.024					68.7
Kapton-C	#0	<350	<13	<14	<11	<6	<51	<2	<3	67.5
	BiPo-3		<0.070		<0.035					67.5
Isotac	#0	<18	<0.45	<0.43	<0.22	<0.18	<2.3	<0.10	<0.14	165.1
Kapton-epoxy	BiPo-3		<0.030		<0.024					95.7

Lowest ^{40}K **activity** quantified (specific activity (8.3±2.4) mBq/kg), with an overall **reduction by a factor ~34**

Recent T-REX publication on improvement of their micromegas

For comparison:
^{40}K specific activity in CYGNO-04 GEMs is 75 mBq/kg

Improvement and assessment of the radiopurity of Micromegas readout planes, J. Castel et al, arXiv:2603.03274

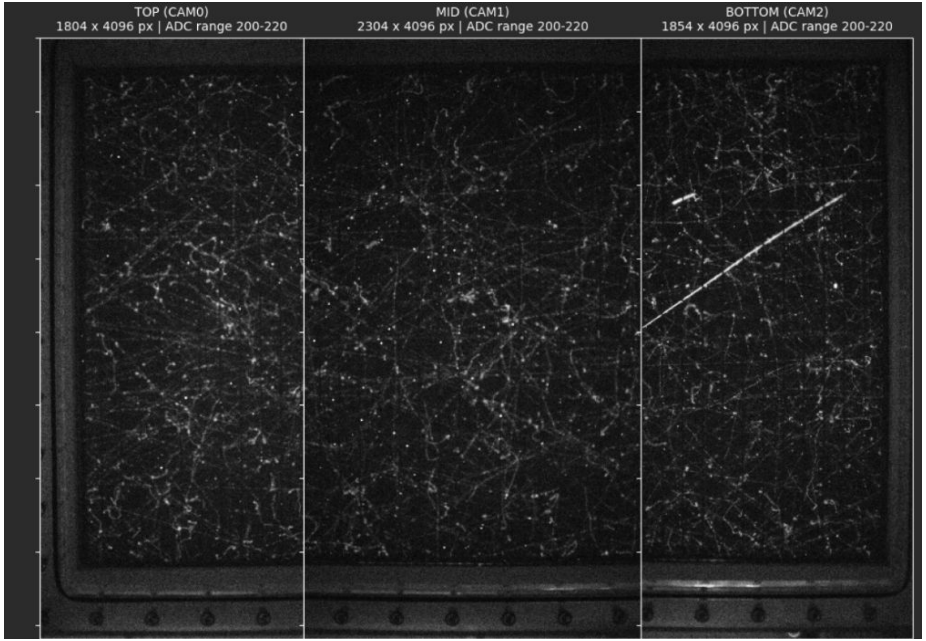
Conclusions

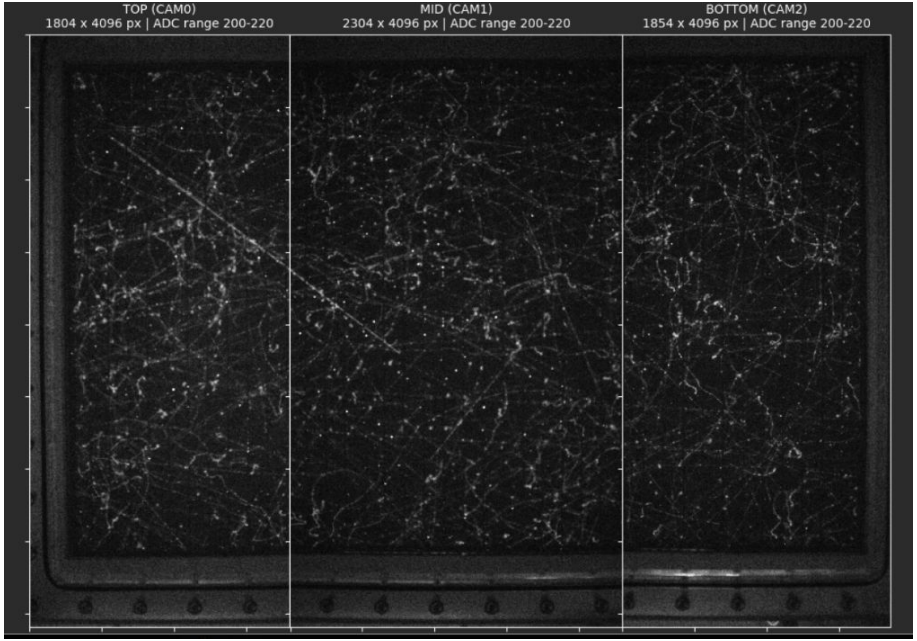
- Rare event searches demand **ultra-low background**
  - mitigation of all background sources: underground facilities, shieldings, fiducialization, particle discrimination
  - **selection of materials**, cleaning and radioactivity screening
- Gaseous TPCs advantages:
  - very good in terms of **spatial and energy resolution, particle identification,** possibility of **directionality**
  - mainly constructed with **radiopure materials** (Cu, kapton, PMMA, nylon) → potentially very low background
  - CYGNO approach with optical readout: cameras/lenses not optimized for low radioactivity. For future detectors: R&D of low background cameras

BACKUP

CYGNO-04 first pictures

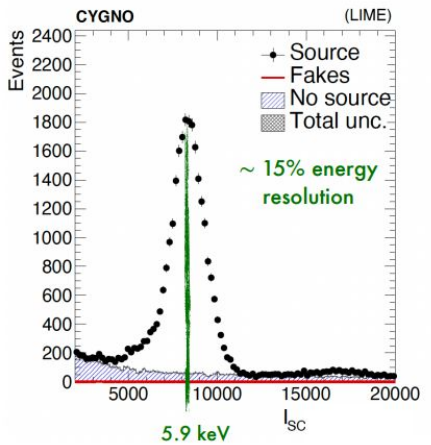
Some of the very first pictures with CYGNO-04 at LNF (overground) August 2026

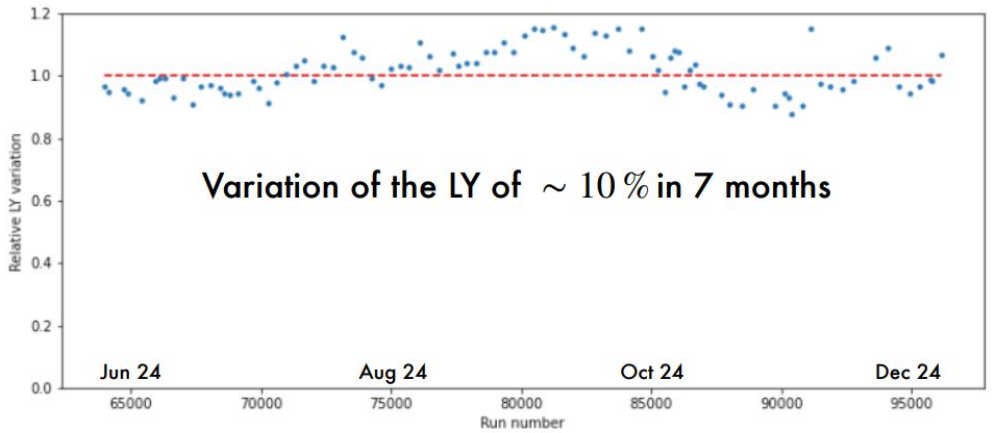




Stability during the underground operation

- **RUN 5:** 7 months of continuous data-taking with no major interruptions
- Light yield measured using a 6 keV ^{55}Fe X-ray source collimated in z



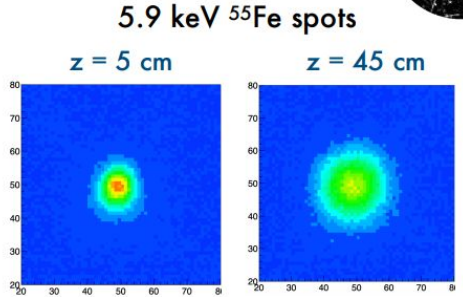


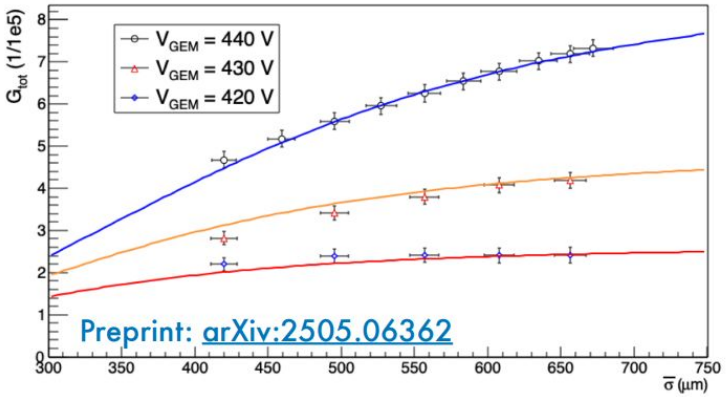
27

S. Piacentini

Saturation of the GEM gain

- **Saturation** of GEM gain due to positive **ions back-flow** screening the electric field, as already found by our CERN colleagues [arXiv:1512.04968]
- **Denser energy deposits ⇒ more charge in the same holes**





We developed a simple model:
[modified Townsend model]

$$\frac{dn}{ds} = \alpha n E(1 - \beta n)$$

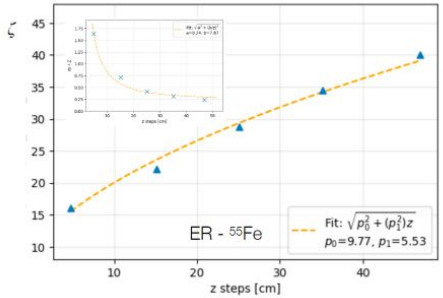
$$G_{tot} = \frac{(ce^{\alpha V_{\mathrm{GEM}}})^3 \sigma^3}{\sigma^3 + (p/V_{\mathrm{GEM}})(ce^{\alpha V_{\mathrm{GEM}}})^2(ce^{\alpha V_{\mathrm{GEM}}} - 1)}$$

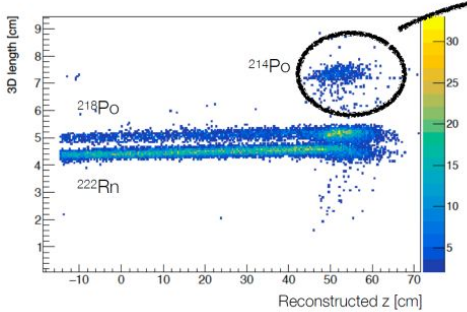
6

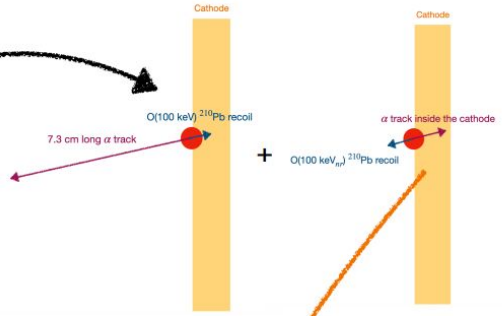
S. Piacentini

z position of alpha tracks

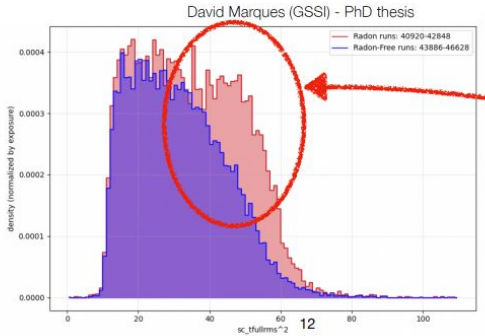
- First application: the **Radon** chain →**alpha tracks**

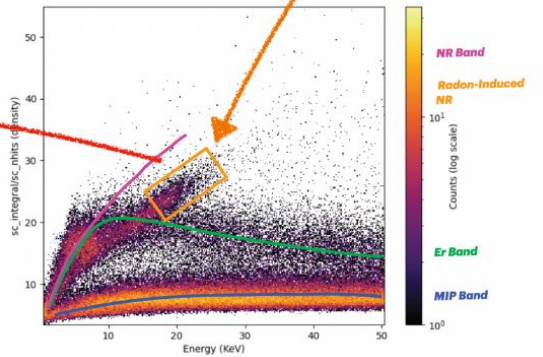






Evidence of an **excess of NR-like events** at the cathode correlated to the activity of Radon progeny in LIME detector

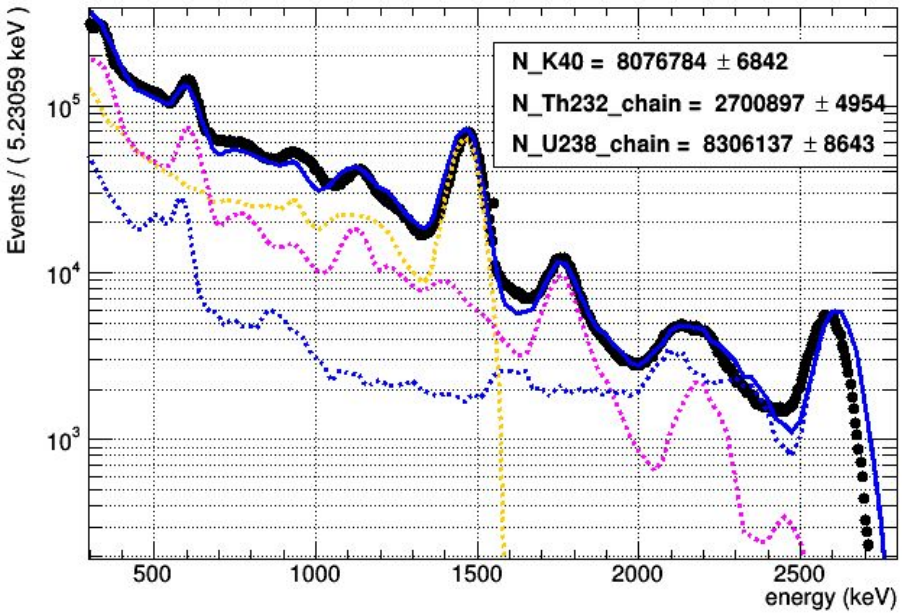




S. Piacentini

Gamma spectrum at LNGS measured with NaI

- Use dataset outside shielding (in LIME control room), ~6 days livetime
- Fit range from 300 keV to 2800 keV



Floating Parameter	FinalValue +/- Error
N_K40	8.0768e+06 +/- 6.84e+03
N_Th232_chain	2.7009e+06 +/- 4.95e+03
N_U238_chain	8.3061e+06 +/- 8.64e+03

Corresponding activities

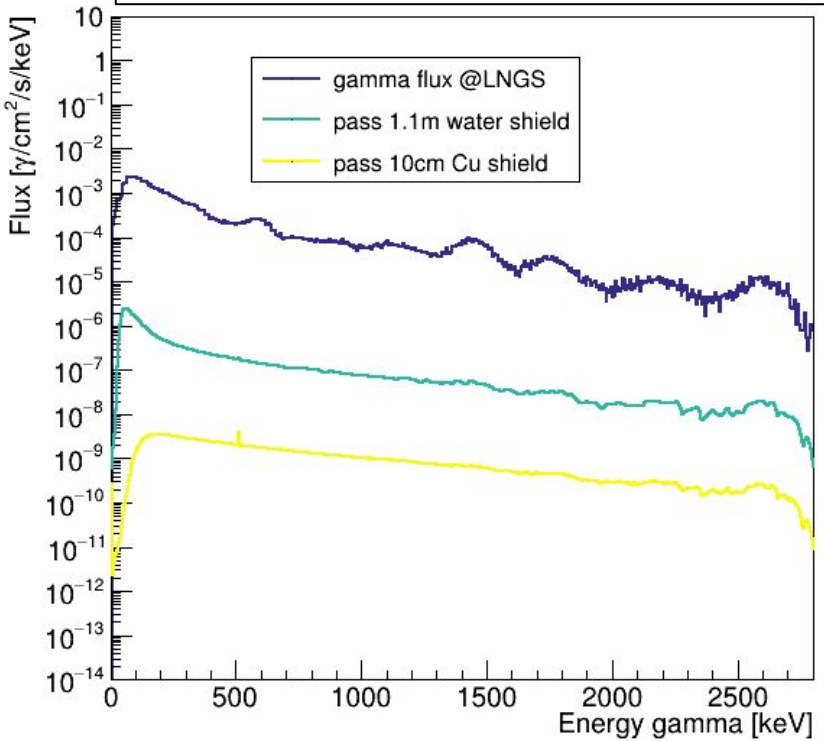
- $^{40}K \rightarrow$ 93.5 Bq/kg
- $^{238}U \rightarrow$ 7.9 Bq/kg
- $^{232}Th \rightarrow$ 4.1 Bq/kg

Counting gammas entering the air sphere

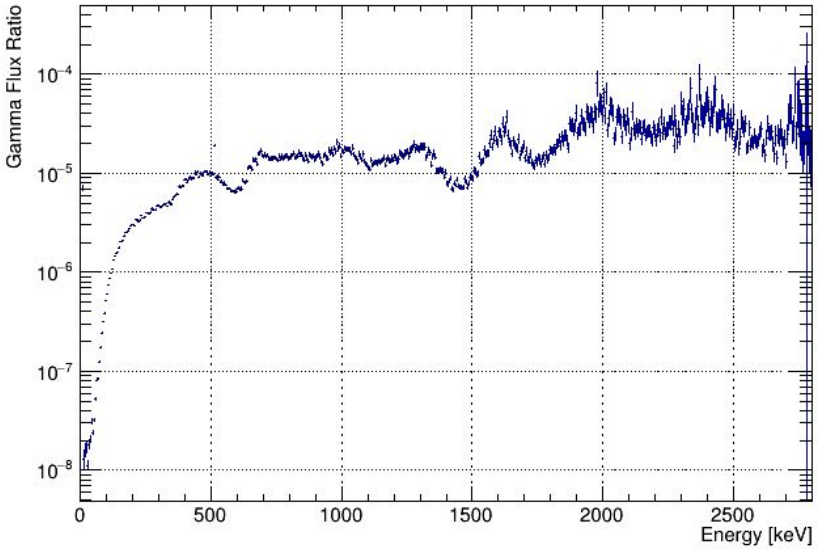
- $^{40}K \rightarrow$ 0.21 gammas/cm^2/s
- $^{238}U \rightarrow$ 0.25 gammas/cm2/s
- $^{232}Th \rightarrow$ 0.12 gammas/cm^2/s
- **Total 0.58 gammas/cm^2/s**

Gamma flux attenuation with CYGNO shielding

gamma flux entering shield: 2.4×10^{-6} cm^{-2}s^{-1}



110 cm water shield
+ 10 cm copper shield



T-REX background

- TREX-DM @LSC underground laboratory
- Two active volumes (each 25 x 25 x 19 cm^3) separated by a central cathode made of mylar inside a 6-cm-thick Cu vessel
- Shielding: 4 cm Cu + 20 cm Pb + 40 cm water
- Operates with 10 bar pressure Ar (Ne) at 10 bar. Active mass 0.32 kg (0.16 kg)
- Expected background between 1 and 10 counts/day/kg/keV

Background assessment for the TREX Dark Matter experiment, J. Castel et al., EPJ-C vol. 79, 782 (2019)

